

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of BSc Biotechnology/ Microbiology/ Biochemistry Examination April-May 2018

Course code & Name: BS402 Basic Molecular Biology-I

04.05.2018

Friday

10:00 AM To 10:20 AM

Maximum Marks: 10

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

10

1. TAT to TAA change is a
 - (i) Missense mutation
 - (ii) Nonsense mutation
 - (iii) Readthrough mutation
 - (iv) Silent mutation
2. During DNA recombination _____ of nucleotides takes place
 - (i) Substitution
 - (ii) Deletion
 - (iii) Rearrangement
 - (iv) Insertion
3. Molecular biology is the study of
 - (i) Carbohydrates and Nucleic acids
 - (ii) Proteins and Lipids
 - (iii) Nucleic acids and Proteins
 - (iv) Lipids and Carbohydrates
4. Select the right trio of protein synthesis
 - (i) Ribosome+mRNA+tRNA
 - (ii) Ribosome+miRNA+mRNA
 - (iii) Ribosome+tRNA+snoRNA
 - (iv) None of the above

5. Solenoid structure is associated with DNA
- | | |
|-------------------|--------------------|
| (i) Repair | (ii) Packaging |
| (iii) Replication | (iv) Recombination |
6. Select the right combination
- | | |
|-------------------------|--------------------------|
| (i) Clover leaf/ rRNA | (ii) Clover leaf/ tRNA |
| (iii) Clover leaf/ mRNA | (iv) Clover leaf/ snoRNA |
7. Nucleic acids by polymerase activity are synthesized in
- | | |
|-----------------------|----------------------|
| (i) 5'→3' direction | (ii) 5'→5' direction |
| (iii) 3'→5' direction | (iv) 3'→3' direction |
8. AUG codes for
- | | |
|---------------|-----------------|
| (i) Alanine | (ii) Methionine |
| (iii) Leucine | (iv) Valine |
9. Select the right combination
- | | |
|----------------------|---------------------|
| (i) APE/ Ribosomes | (ii) APC/ Ribosomes |
| (iii) APD/ Ribosomes | (iv) ABD/ Ribosomes |
10. Poly (A) nucleotide chain is found at the 3'end of
- | | |
|------------|-----------------------|
| (i) rRNA | (ii) tRNA |
| (iii) mRNA | (iv) all types of RNA |

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IV Semester of BSc Biotechnology/ Microbiology/ Biochemistry Examination April-May 2018

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Friday

10:21 AM To 12 Noon

Maximum Marks: 25

Instructions:

1. Section I and II must be attempted in SINGLE ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

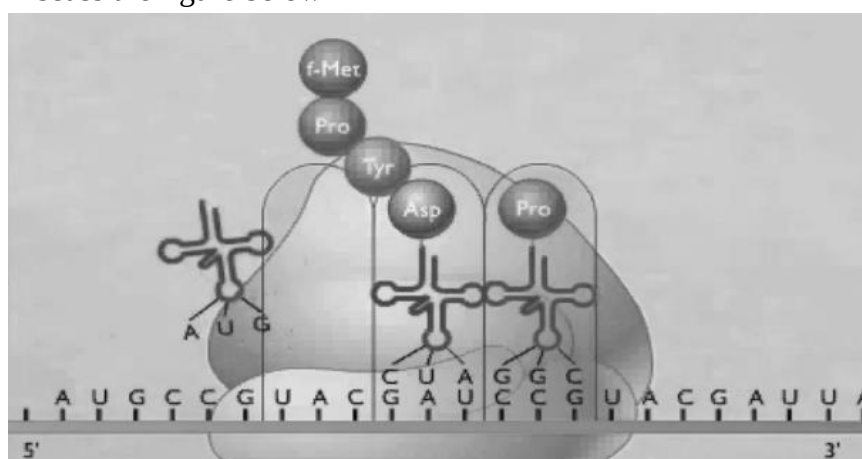
SECTION - I

Q - II Answer the following questions as directed

10

1. Discuss the figure below

5



OR

Compare “genome organization” between prokaryotes and eukaryotes

2. Describe transcription in **ANY ONE**
(i) Prokaryotes (ii) Nucleus possessing unicellular organisms

5

SECTION – II

Q-III Answer the following questions as directed

15

1. Describe the experiment that used various enzymes to proof that DNA is genetic material

5

OR

Which nitrogenous base in DNA gets affected due to persistent UV exposure to cells? What is it known as and how this damage is taken care by the cell? Also, describe very briefly various other mechanisms that restore different types of DNA damage.

2. Describe genome replication briefly (**ANY ONE**)
(i) Ecoli (ii) Yeast
3. Describe **ANY ONE**
(i) Rolling circle mechanism (ii) Gene mutation

5

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